

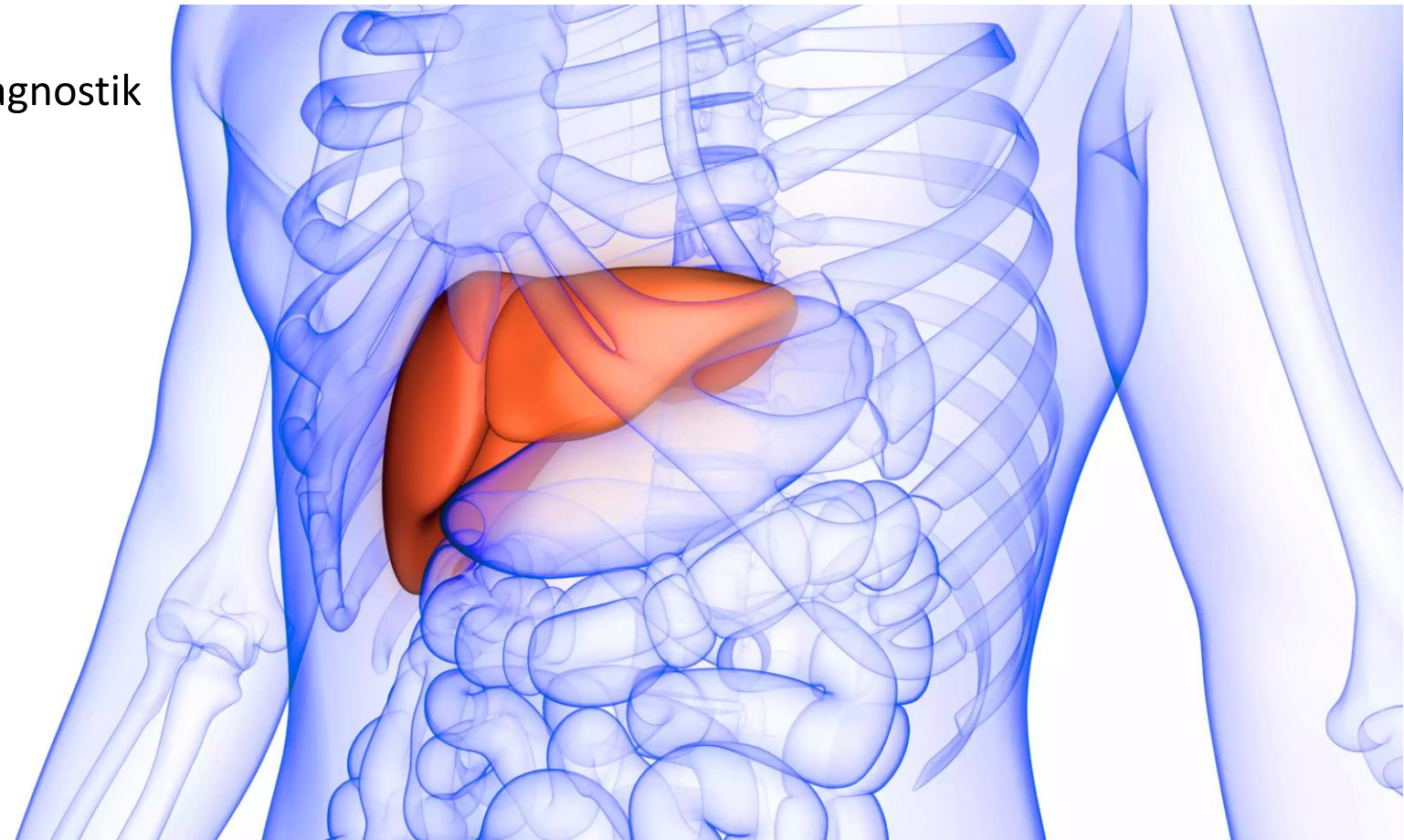
Pathophysiology of liver disease

Fordøjelsessystem, patofysiologi og diagnostik

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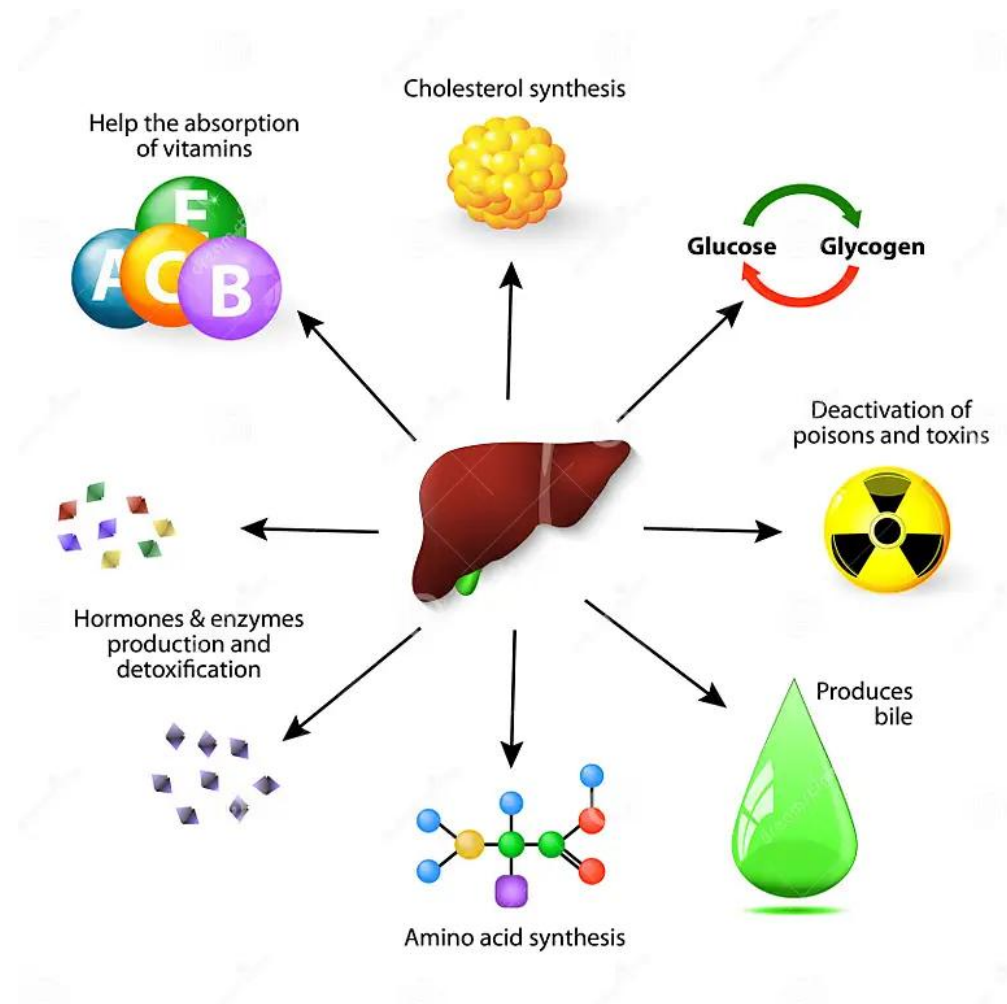
Læringsmål

- *Med baggrund i de patofysiologiske forhold, beskriv symptomer og objektive fund som kan ses hos en leverpatient*
- *Beskriv de kliniske konsekvenser af leversvigt*
- *Beskriv patogenesen af lever cirrose*
- *Gør rede for serologiske markører for leversygdom (levertal)*
- *Beskriv de vigtigste årsager til akut og kronisk hepatitis (herunder viral, autoimmun, medikament-, og alkohol-induceret hepatitis)*

Liver: function

Four main physiological roles

- Metabolic
- Excretory
- Protective/detoxification
- Synthetic



Liver: normal histology

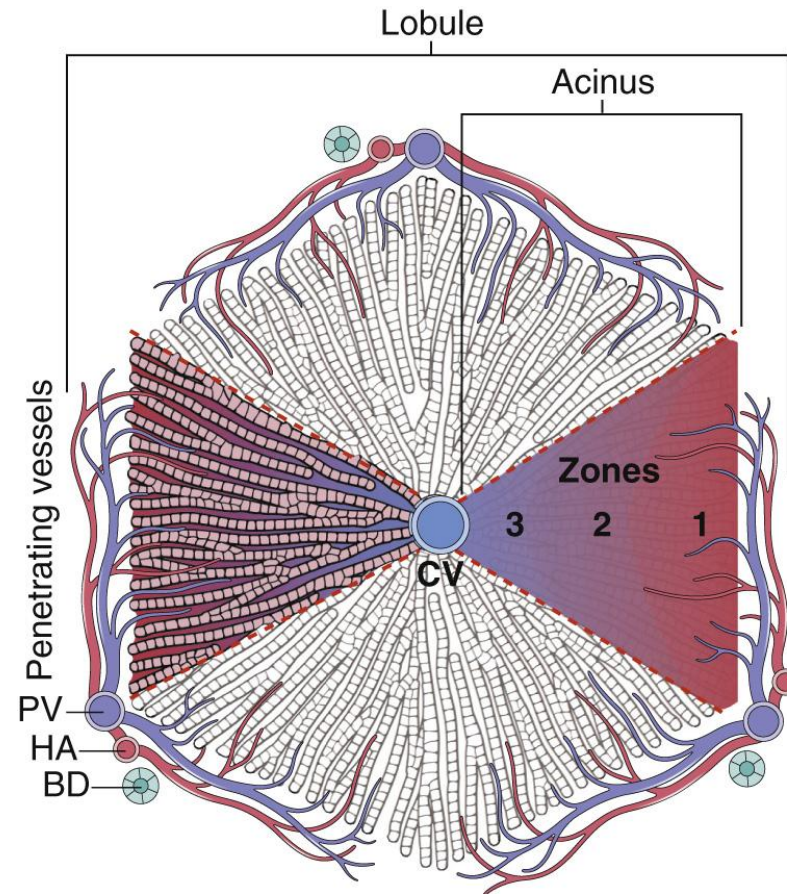
Lobular model

- The terminal hepatic vein is at the center of a “lobule”
- The portal tracts are at the periphery.
- Pathologists often refer to the regions of the parenchyma as “periportal and centrilobular.”

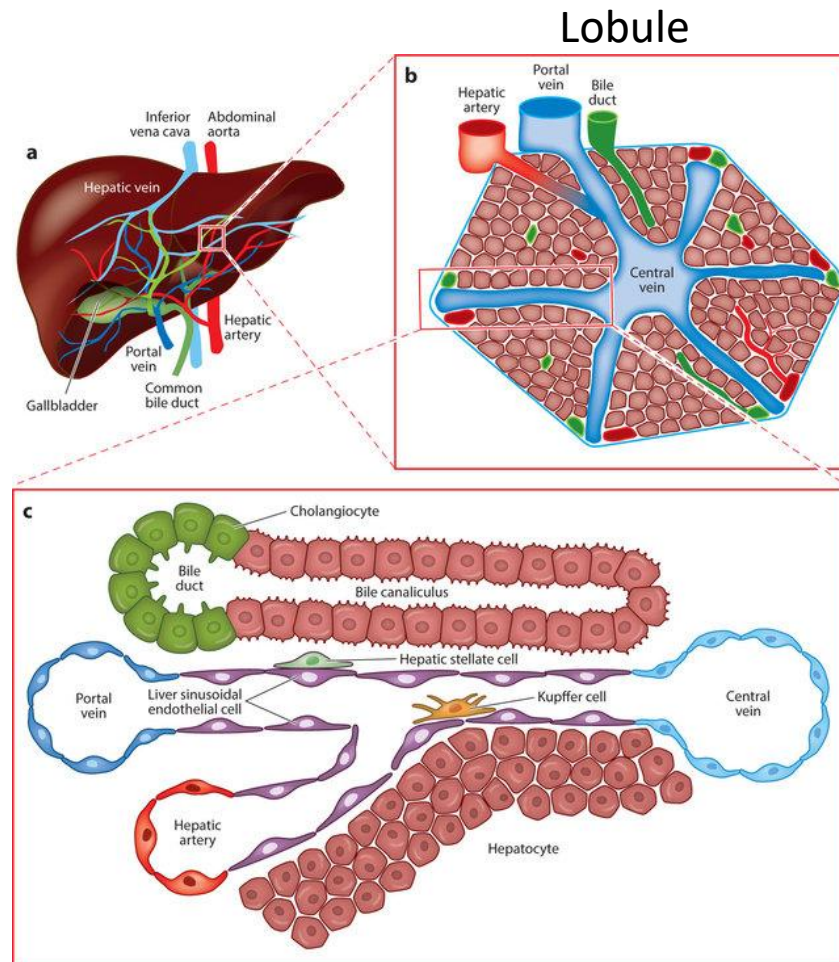
Acinar model

- On the basis of blood flow
- Three zones
 - zone 1 -> closest to the blood supply
 - zone 3 -> farthest to the blood supply

Models of liver anatomy



Liver: circulation of blood and bile



- The liver receives blood from:
 - **V portae** (~75 %) from the gut - nutrients and other substances
 - **A hepatica** (~25 %) oxygenated blood
- The two blood types are mixed in the **sinusoids** and circulate to the **centrral vein**
- Blood abandons the liver via the liver vein to the **V cava inferior**
- The bile transports **bilirrubin**, **cholesterol** and **toxic byproducts** out of the liver

Liver disease: pathophysiological basis

- **Liver disease:** any condition where tissue injury affects the normal function of the liver
- **Causes of hepatic tissue injury:**
 - Infectious, metabolic, autoimmune, and genetic factors
 - Obstruction to blood vessels or bile ducts
 - Cancer

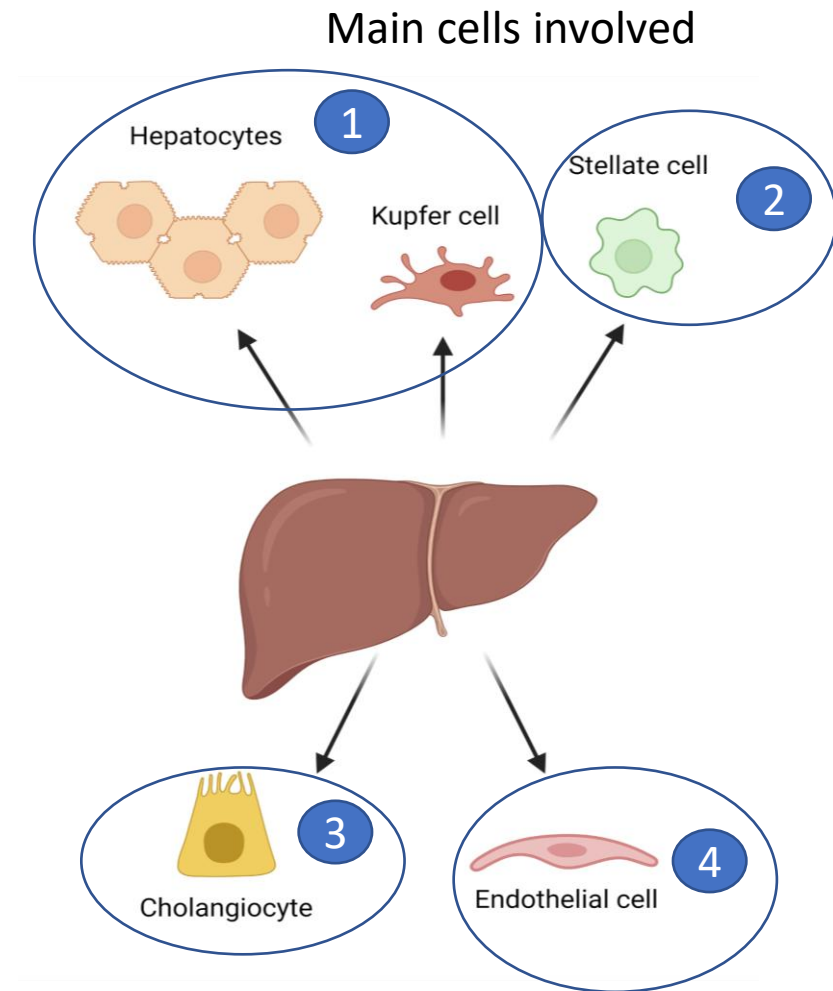


Patterns of hepatic tissue injury

1. Hepatocyte injury
 - Cell death (necrosis or apoptosis)
 - Degenerative and/or intracellular accumulations (e.g., iron, fat)
 - Inflammation
2. Stellate cell activation
 - Fibrosis -> cirrhosis
3. Biliary injury (cholestatic syndromes)
4. Circulatory disorders

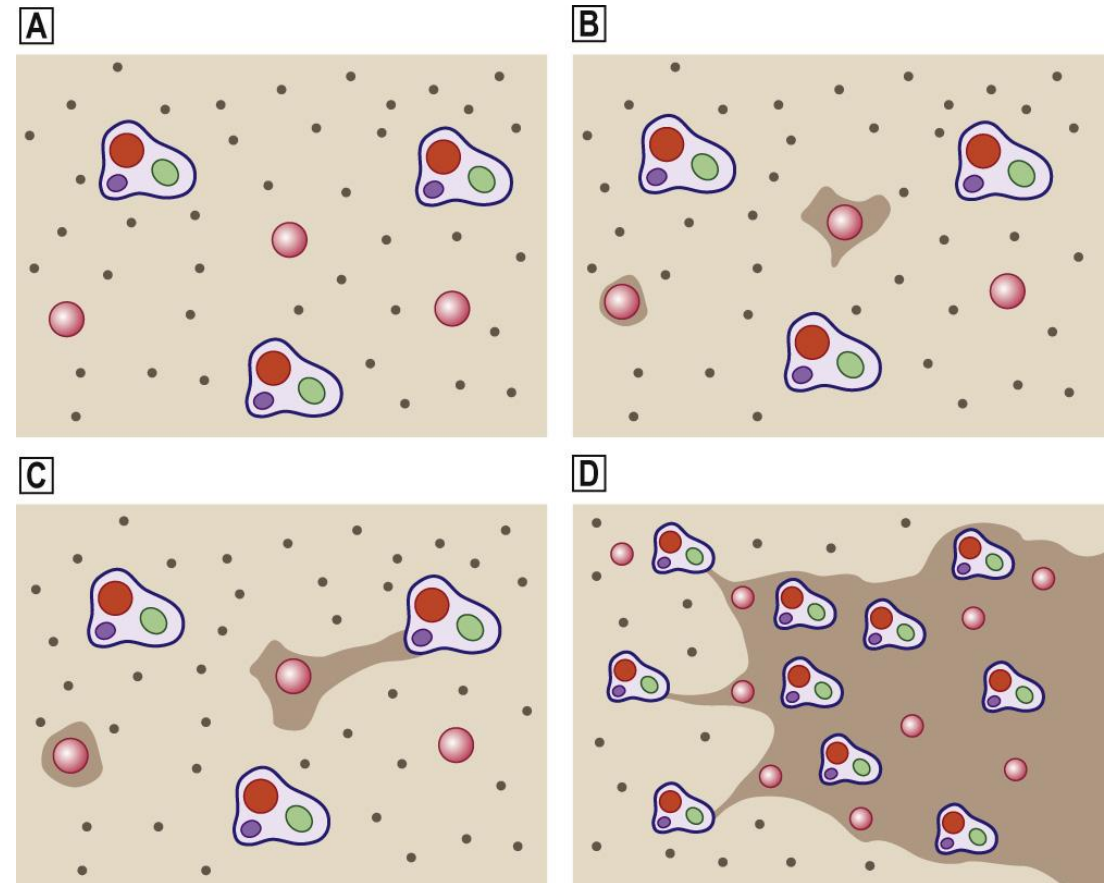
The clinical consequences are highly dependent on severity of injury, and effectiveness of regeneration

Liver disease is classified as acute or chronic



Acute liver disease

- Damage resolves within 6 months
- Histological features
 - A. Hepatocyte apoptosis (dots)
 - B. Death of a confluent zone of hepatocytes (zonal necrosis)
 - C. More extensive necrosis in a band pattern (bridging necrosis)
 - D. Severe hepatitis (multiacinar necrosis)
- Common etiology:
 - Viral hepatitis
 - Drug/alcohol overdose

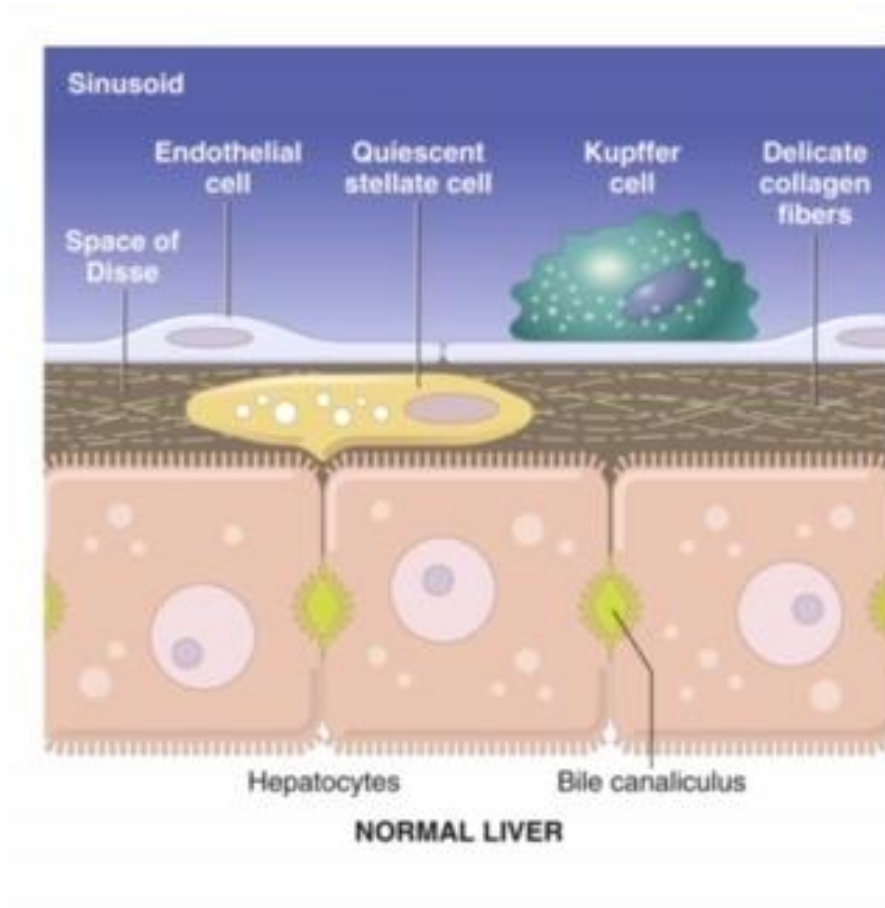


Chronic liver disease

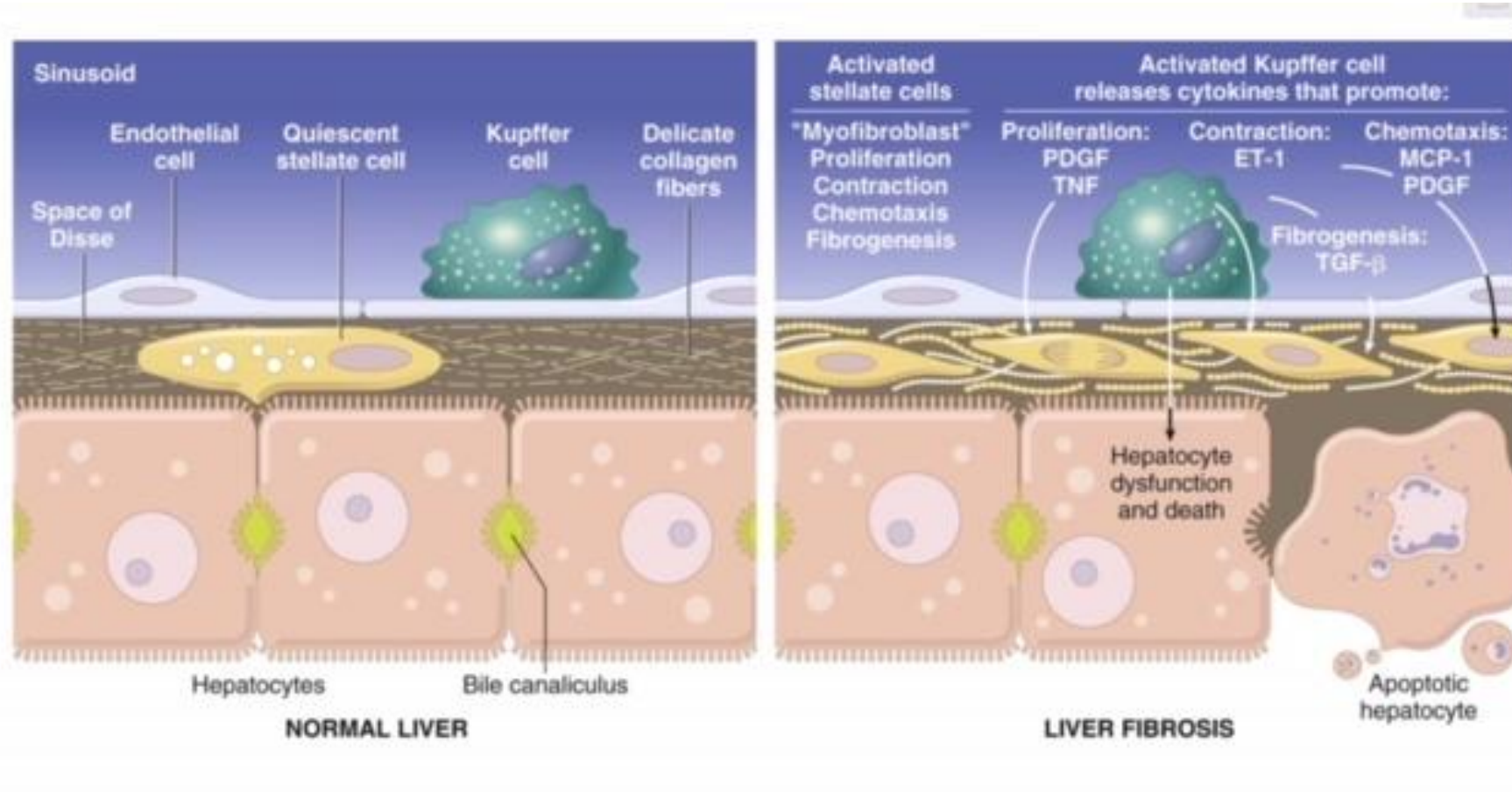
- **Progressive deterioration** of liver functions (> 6 months)
- Histological features:
 - Continuous process of inflammation, destruction, and regeneration of liver parenchyma, which leads to fibrosis and eventually to cirrhosis.
- Common etiology
 - Viral/autoimmune hepatitis
 - Fatty liver disease (alcoholic/metabolic)
 - Cholestasis
 - Genetic diseases
 - Ischemia (e.g. thrombosis of hepatic veins)



Liver fibrosis

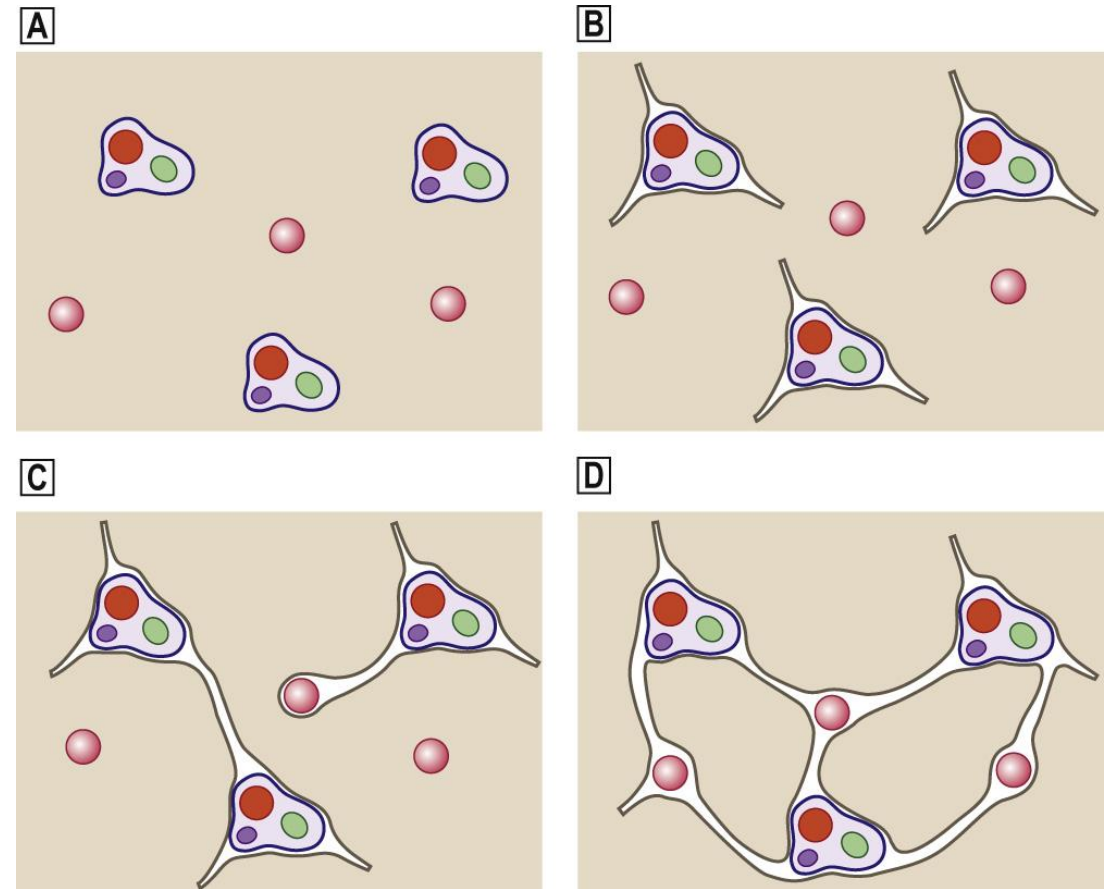


Liver fibrosis



Cirrhosis

- Diffuse transformation of the liver into regenerative parenchymal nodules surrounded by fibrous bands
- Evolution:
 - A. Normal vascular relationships
 - B. Fibrotic expansion
 - C. 'Bridging fibrosis' between portal tracts and/or hepatic veins
 - D. Islands of regenerating hepatocytes enveloped by fibrous scar tissue (end-stage cirrhosis)

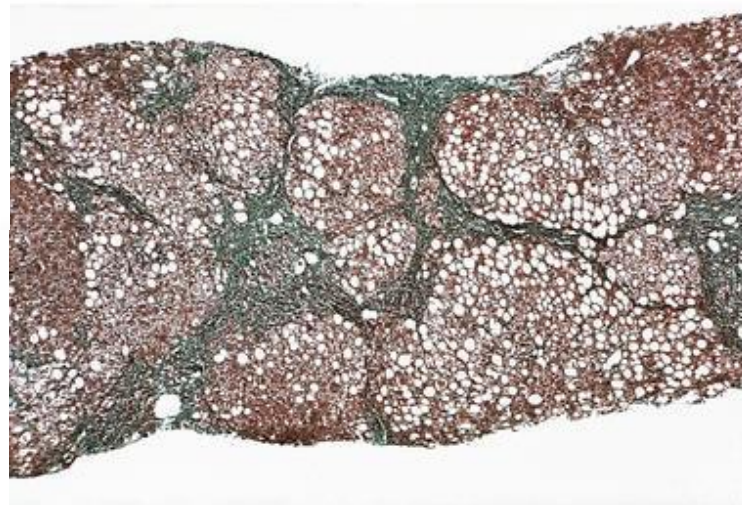


Cirrhosis

Liver with micronodular cirrhosis



Regeneration nodules (ca. 2 mm in diameter)



Histology revealing nodules of steatotic hepatocytes surrounded by dense connective tissue (collagen stain in green)

Liver failure

Definition:

When the liver can no longer function sufficiently to maintain the body's homeostasis

- Inadequate synthesis (albumin, clotting factors, glucose from glycogen)
- Inadequate metabolism and elimination of endogenous products (bilirubin, nitrogenous waste, hormones, etc)

Clinical presentation

- **Acute liver failure** — acute severe liver cell injury with insufficient regeneration (massive necrosis)
- **Chronic liver failure** — decompensation of liver function, occurs late during chronic liver disease (typically **cirrhosis**)

Signs and symptoms of liver disease

Typical of:
-Acute
-Chronic



Fatigue

Hepatic encephalopathy

← Jaundice (icterus)

Haematemesis →



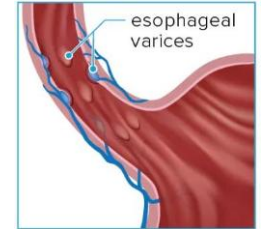
Loss of appetite

Portal hypertension



← Spider naevi

Esophageal varices →



Abnormal bleeding

Weakness



← Gynaecomastia

Oedema →



Cholestasis

Pruritus



← Ascites

Bruising →



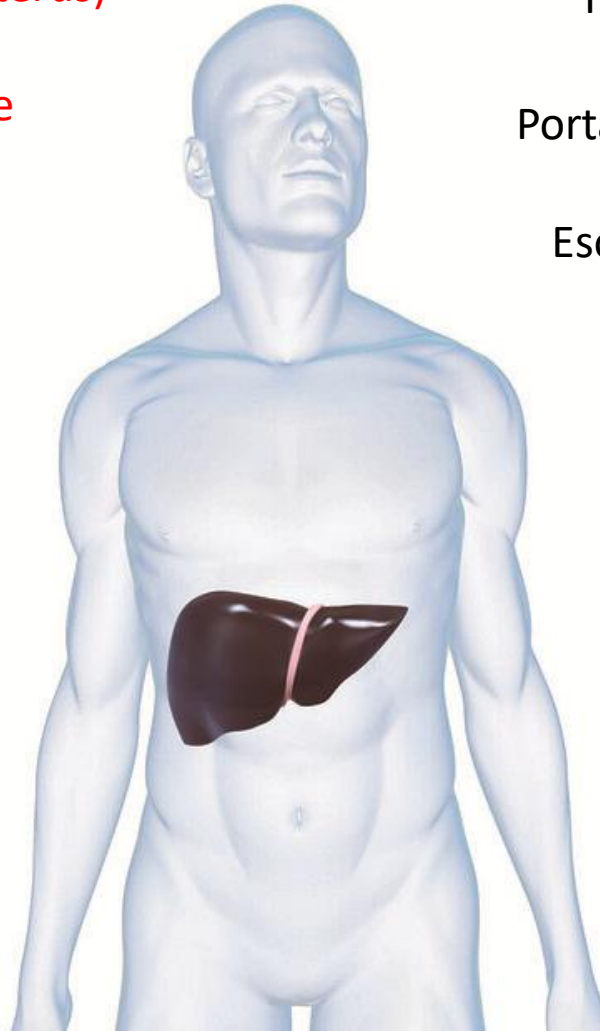
Hepatomegaly

Liver failure

Steatorrhoea

Dark urine →

Pale faeces



Pathological basis of symptoms and signs of liver disease

Sign or symptom	Pathological basis
Jaundice	• Impaired conjugation and/or excretion of bile • Biliary obstruction
Hepatomegaly	• Increased size of liver due to inflammation (e.g. hepatitis), infiltration (e.g. amyloid, fat) or tumour (primary or secondary)
Dark urine	• Conjugated hyperbilirubinaemia due to failure of bile excretion
Pale faeces	• Bile duct obstruction causing lack of bile pigments in faeces
Spider naevi, Gynaecomastia	• Secondary to hyperoestrogenism, failure of hormone metabolism
Oedema	• Reduced plasma oncotic pressure due to hypoalbuminaemia
Steatorrhoea	• Malabsorption of fat due to lack of bile (e.g. biliary obstruction)
Pruritus	• Biliary obstruction resulting in bile salt accumulation
Ascites	• Combination of hypoalbuminaemia, portal hypertension and secondary hyperaldosteronism
Bruising or bleeding	• Impaired hepatic synthesis of clotting factors
Splenomegaly	• Enlargement of spleen due to congestion in portal hypertension
Haematemesis	• Ruptured oesophageal varices due to portal hypertension
Encephalopathy	• Failure of liver to remove exogenous or endogenous substances mimicking or altering balance of neurotransmitters

Investigation of liver disease

- Laboratory evaluation
 - Serum liver enzymes (ASAT & ALAT) -> hepatocyte integrity
 - Serum albumin | -> hepatocyte synthetic function
 - Clotting factors |
 - Serum bilirubin | -> liver excretory capacity
 - Serum alkaline phosphatase / GGT |
 - Serum ammonia -> hepatocyte metabolism
 - Immunology (specific IgGs)
 - Virology
- Imaging
- Biopsy and histopathology

To be continued....

Summary of acute vs. chronic liver disease

	Acute	Chronic
Duration	<6 months	
Histological changes	• Hepatocyte ballooning • Necrosis/apoptosis of hepatocytes • Centroacinar necrosis • Parenchymal mononuclear infiltration	
Symptoms / signs	• General discomfort, loss of appetite, jaundice, enlarged liver	
Most common etiology	• Viral hepatitis • Drug/alcohol overdose	
Clinical consequences	• Resolution • Chronic hepatitis • Liver failure	

Summary of acute vs. chronic liver disease

	Acute	Chronic
Duration	<6 months	>6 months
Histological changes	• Hepatocyte ballooning • Necrosis/apoptosis of hepatocytes • Centroacinar necrosis • Parenchymal mononuclear infiltration	• Periportal inflammation • Bridging necrosis • Scarring (fibrosis) • Fibrous septa -> nodules
Symptoms / signs	• General discomfort, loss of appetite, jaundice, enlarged liver	• Weakness, jaundice, ascites, bruising, encephalopathy, bleeding from oesophageal varices
Most common etiology	• Viral hepatitis • Drug/alcohol overdose	• Viral hepatitis • Drug/alcohol induced • Autoimmune • Metabolic (non-alcoholic)
Clinical consequences	• Resolution • Chronic hepatitis • Liver failure	• Steatosis • Cirrhosis • Liver failure

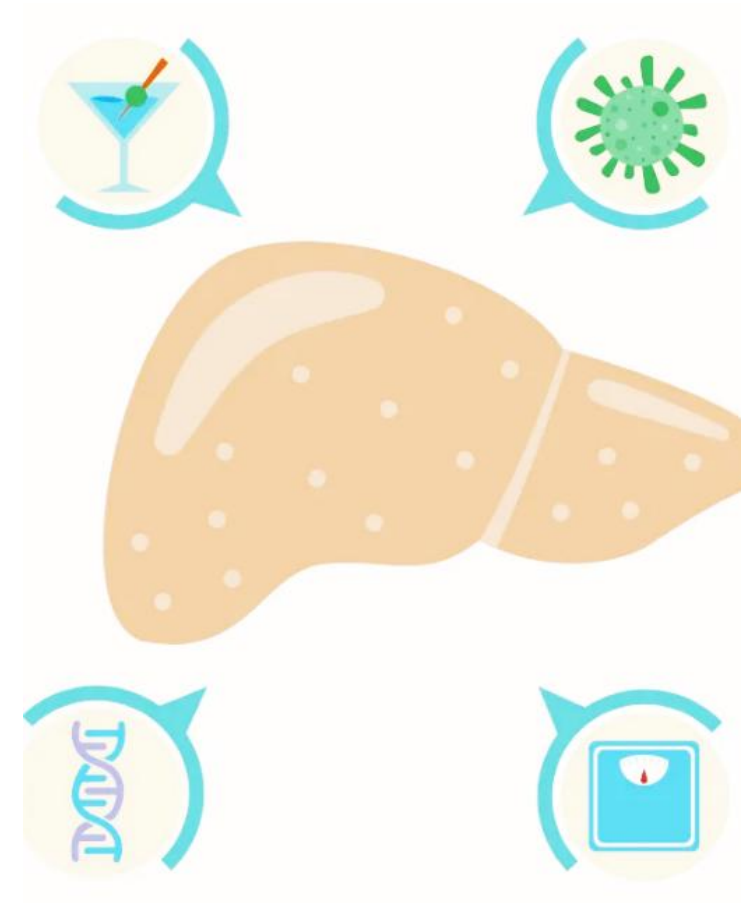
Etiology of liver disease

Liver injury: Infectious, metabolic, environmental, autoimmune, and genetic factors

Etiology of liver disease

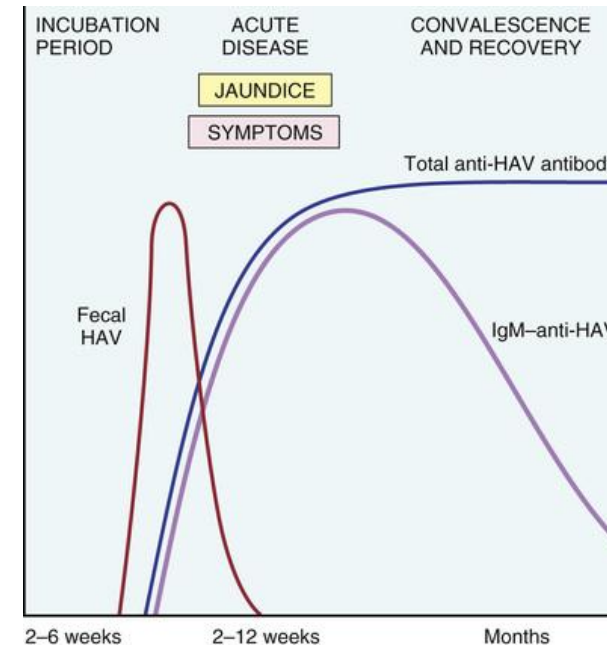
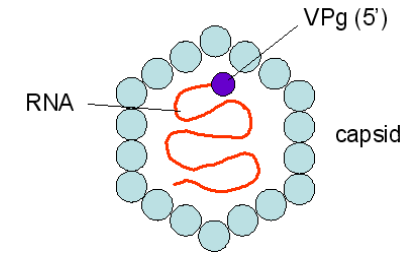
The causes of liver disease include

- Infectious
- Metabolic
- Environmental
- Autoimmune
- Genetic factors

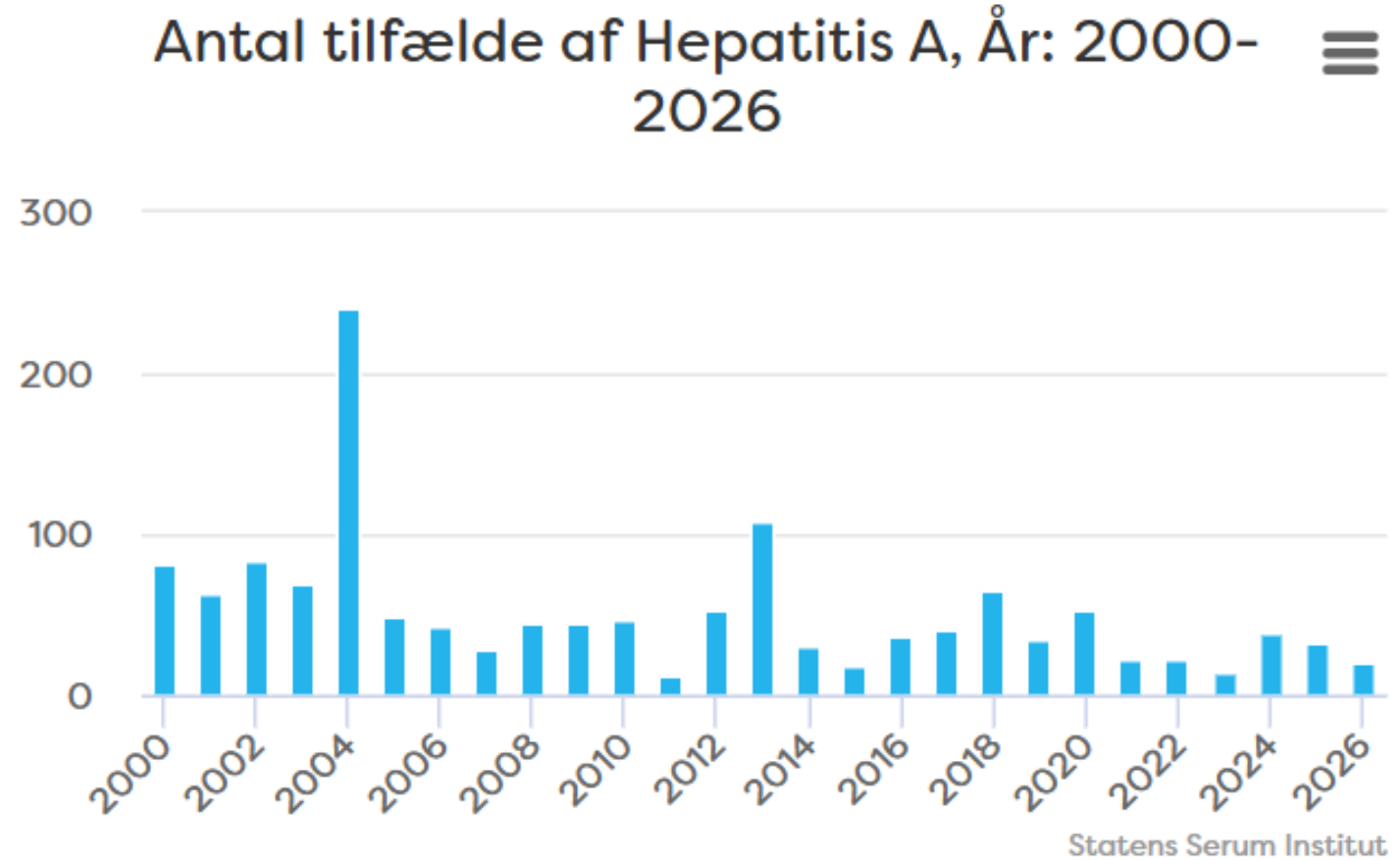
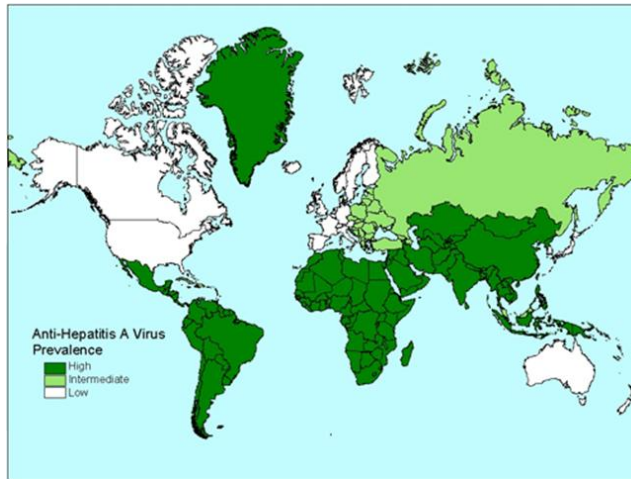


Hepatitis A

- HAV: ssRNA, non enveloped virus
- Symptoms: Anorexia, nausea, and jaundice. Occasionally subclinical
- Frequency of chronic liver disease: never (typically a benign, self-limited disease)
- Route of transmission: Fecal-oral
- Incubation time: 2-6 weeks
- Pathophysiology: acute lobular hepatitis +/- cholestasis

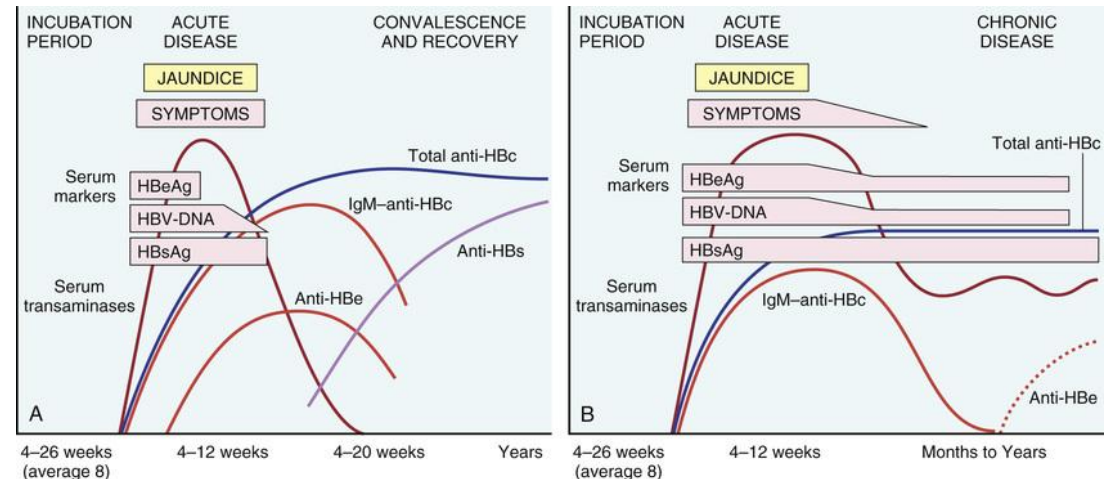
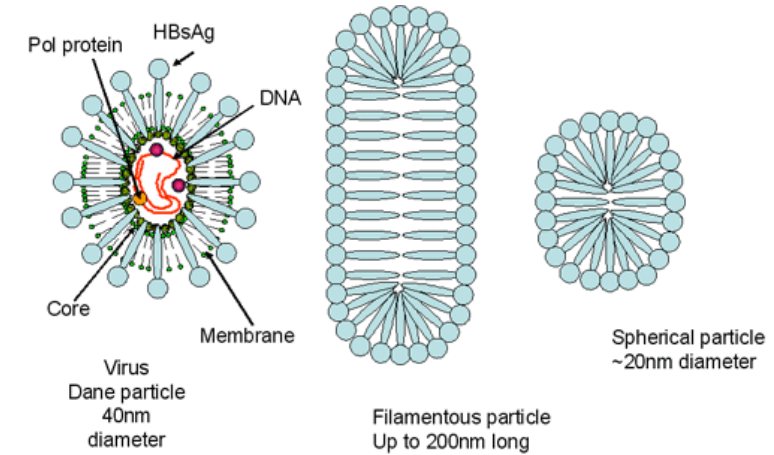


Hepatitis A

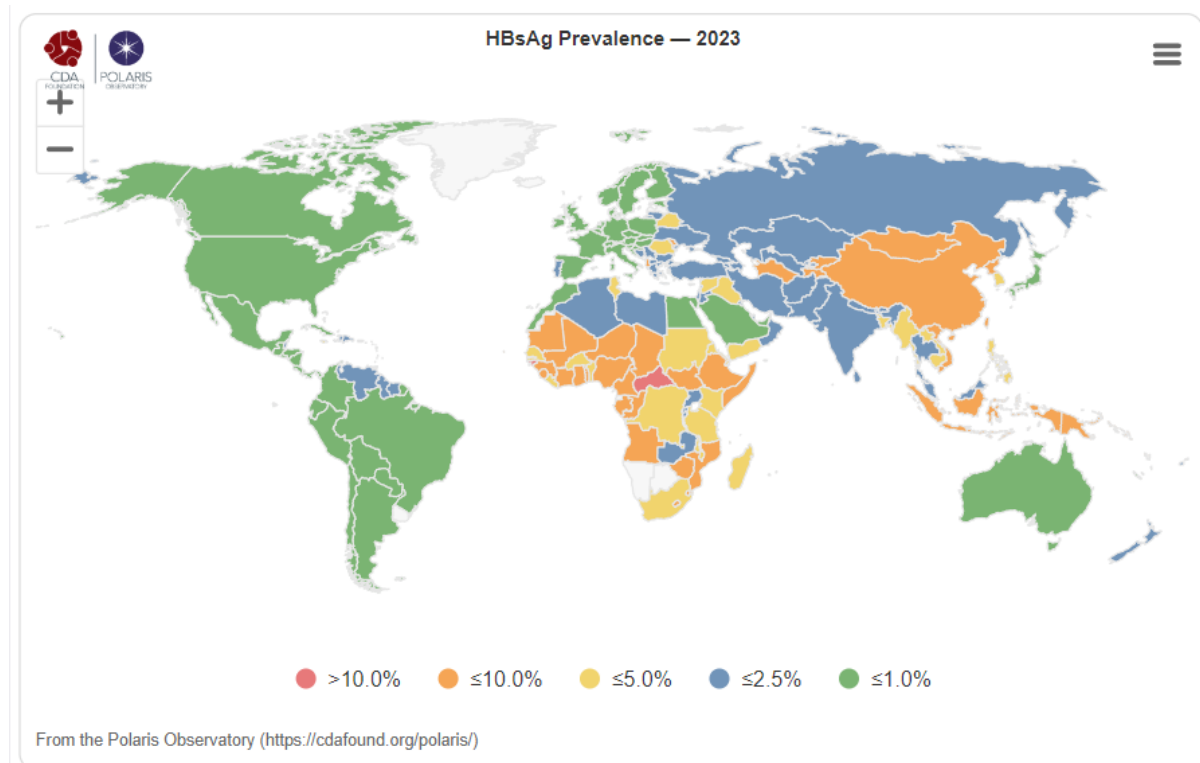


Hepatitis B

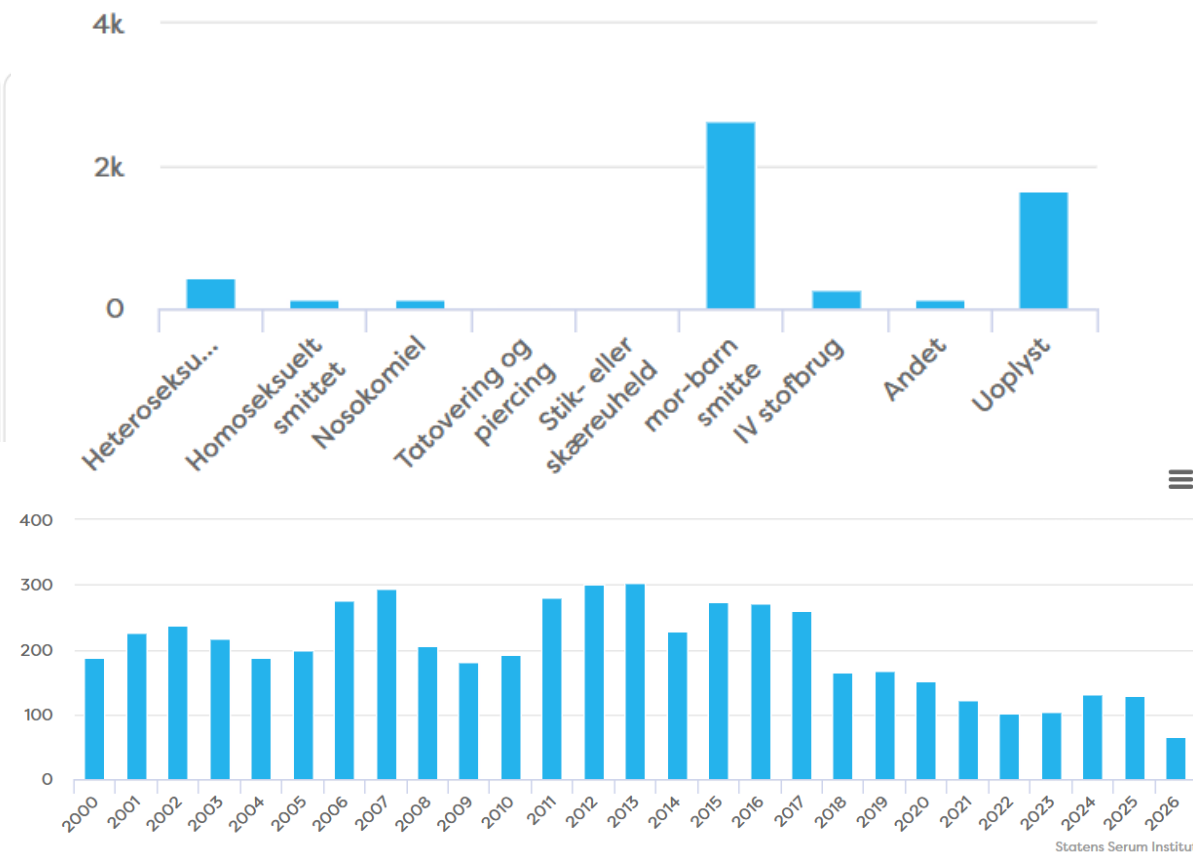
- HBV: dsDNA, enveloped infectious (core antigen HBcAg, surface antigen HBsAg) / non-infectious virus particles
- Symptoms: Often subclinical disease (“healthy” carrier state) but can give acute symptoms
- Frequency of chronic liver disease: 5-90%
- Route of transmission: Parenteral, sexual contact, perinatal, needle stick
- Incubation time: 4-26 weeks
- Pathophysiology: Histologic features of acute and chronic hepatitis
- Chronic hepatitis is associated with risk of hepatocellular carcinoma



Hepatitis B

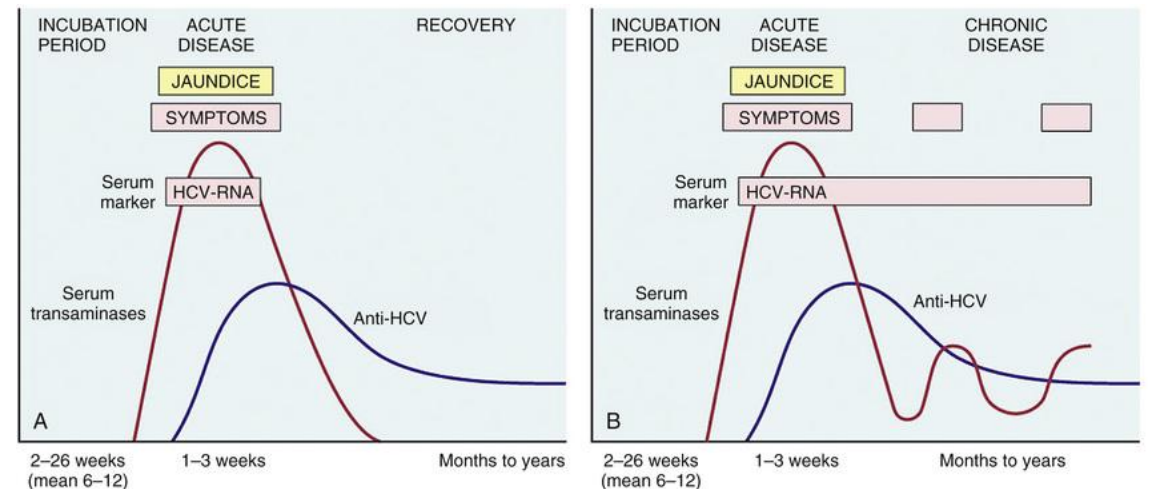
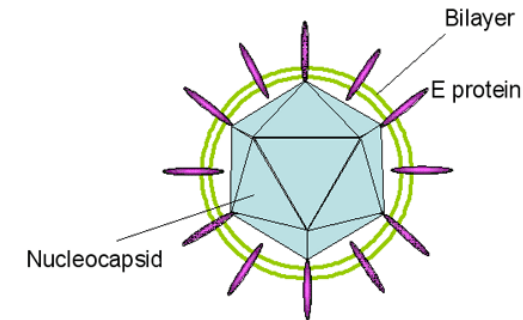


Antal tilfælde af Hepatitis B, År: 2000-2026

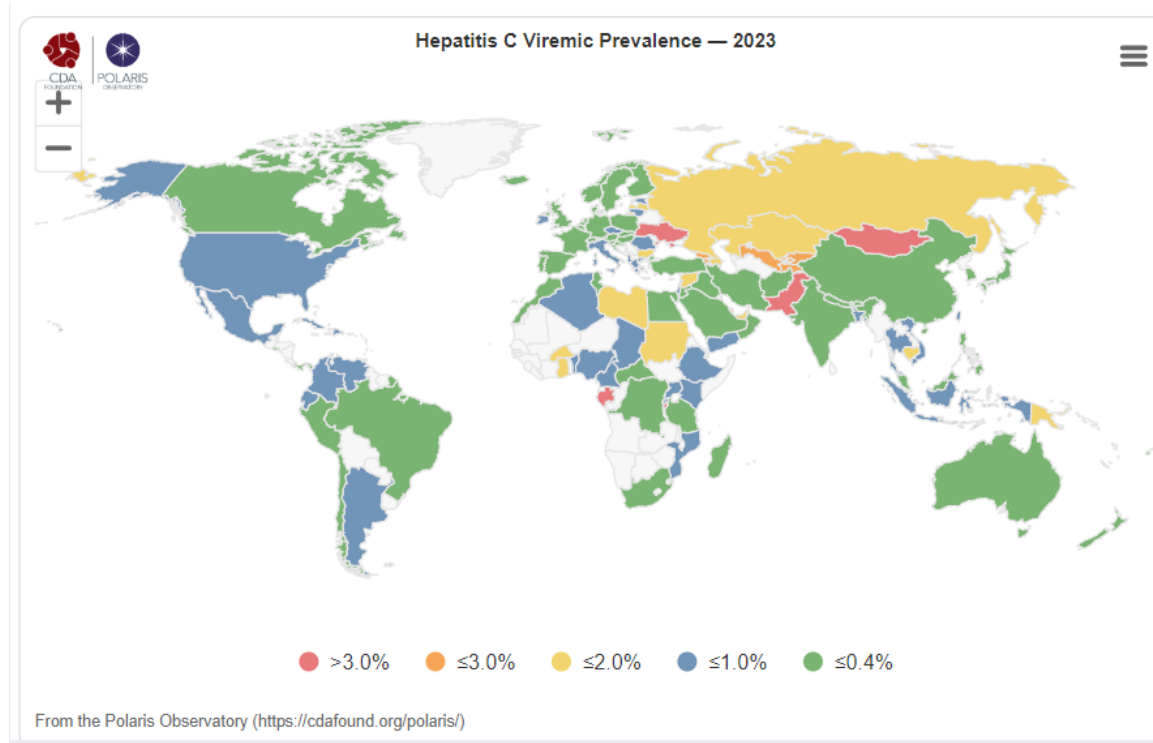


Hepatitis C

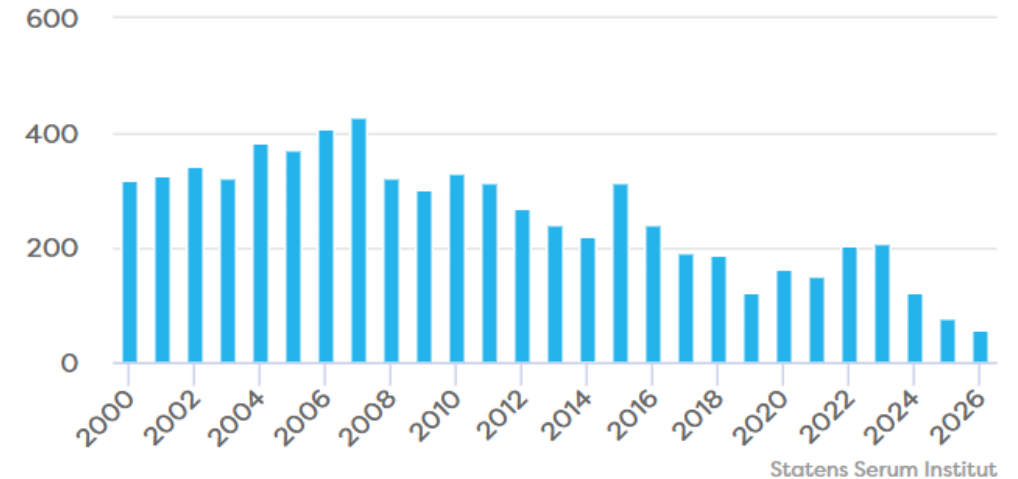
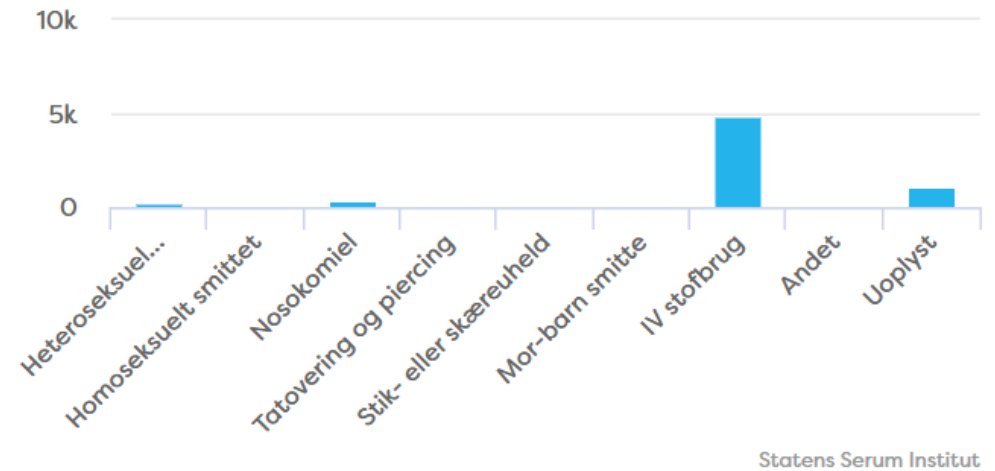
- HCV: Enveloped ssRNA virus, **various genotypes**
- Symptoms: Often asymptomatic in the acute phase (75%)
- Frequency of chronic liver disease: 80%
- Route of transmission: Parenteral, needle stick, sexual contact
- Incubation time: 2-26 weeks
- Pathophysiology:
 - Portal, interface and lobular hepatitis
 - Progressive fibrosis –cirrhosis
 - Variable steatosis



Hepatitis C



Antal tilfælde af Hepatitis C, År: 2000-2026

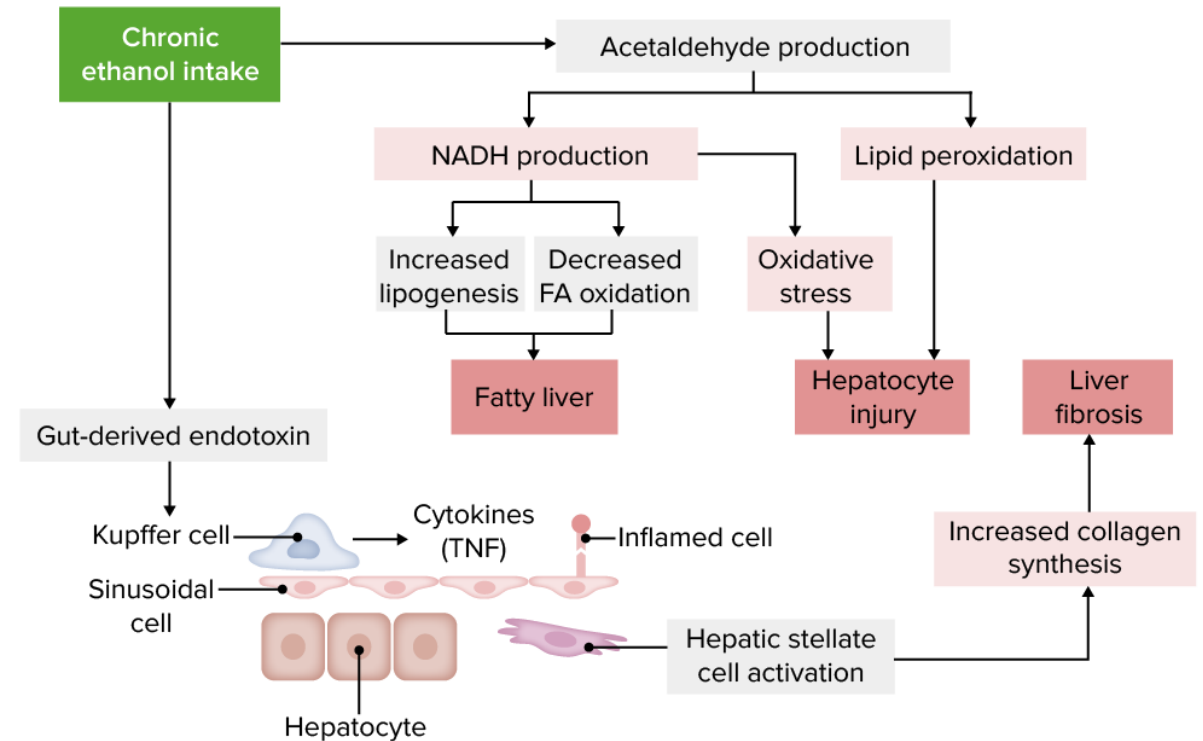


Oversigt over infektion med hepatitis

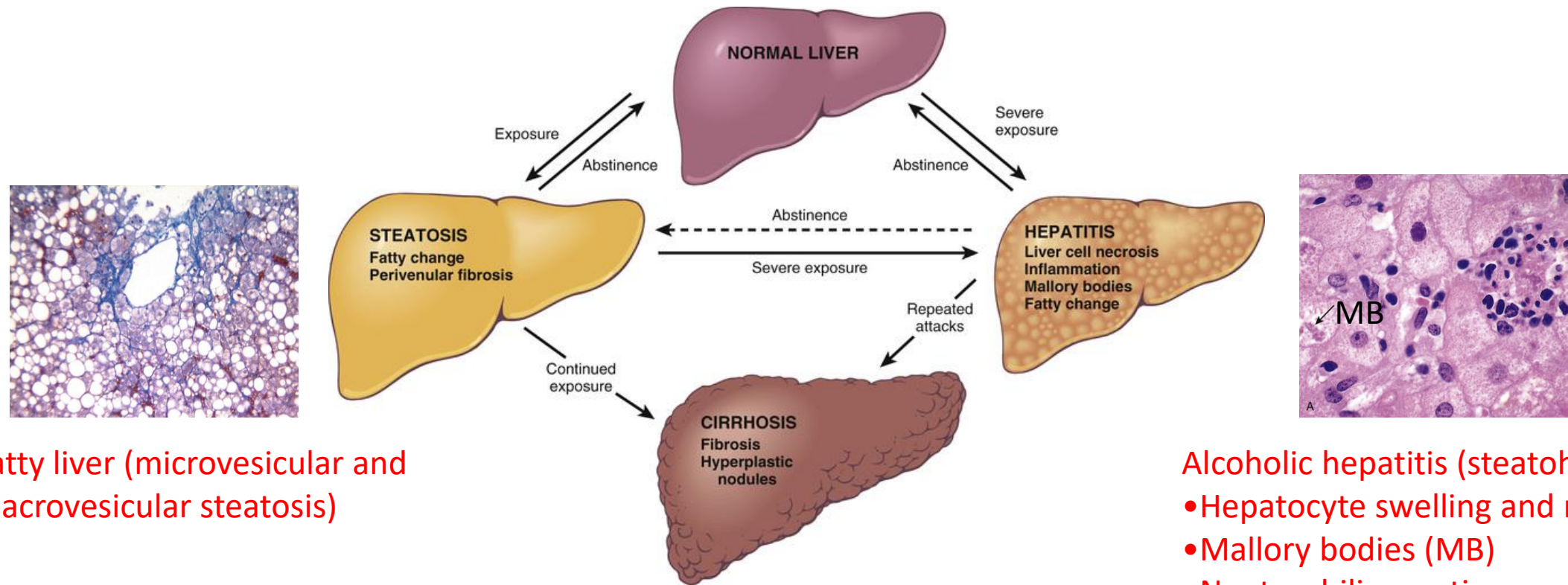
Egenskab	Hepatitis A virus (HAV)	Hepatitis B virus (HBV)	Hepatitis C virus (HCV)
Virustype	Picornavirus	Hepadnavirus	Flavivirus
Genom	RNA	DNA	RNA
Smitemåde/-vej			
• Blod	Nej	Ja	Ja
• Seksuel	Sjælden	Ja	Sjælden
• Mor til barn	Sjælden	Ja	Sjælden
• Husstandskontakt	Ja (fækal-oral)	Sjælden	Nej
Estimeret antal kronisk inficerede i Danmark	Ingen kroniske tilfælde	Ca. 15.000	Ca. 17.000
Estimeret antal kronisk inficerede globalt	Ingen kroniske tilfælde	Ca. 254–350 millioner	Ca. 50–170 millioner
Hyppigste smitemåde for danskere i Danmark	Fødevarer og rejser	Seksuel kontakt	Intravenøst stofmisbrug
Hyppigste smitemåde for udlændinge i Danmark	Rejser til endemiske områder	Mor til barn	Ikke-screenet blod eller stofmisbrug
Hyppigste smitemåde globalt	Fækal-oral via kontamineret vand/mad	Mor til barn	Nosokomial smitte og stofmisbrug
Inkubationstid	2–6 uger	4–26 uger	2–26 uger
Akut infektion	Ofte med symptomer	Ofte asymptomatisk (børn), symptomer (voksne)	Ofte asymptomatisk i akutfasen
Kronisk bærertilstand	Nej	5–90% afhængig af alder ved smitte	50–80%
Beskyttende immunitet efter overstået infektion	Ja, livslang	Ja, livslang	Nej
Vaccine	Ja	Ja	Nej

Alcoholic liver disease

- Caused by chronic ethanol consumption
- Leading cause of liver disease
- ~8-20% eventually develop cirrhosis
- Several mechanisms of tissue injury
- Severity depends on genetic, metabolic and immunologic factors



Alcoholic liver disease

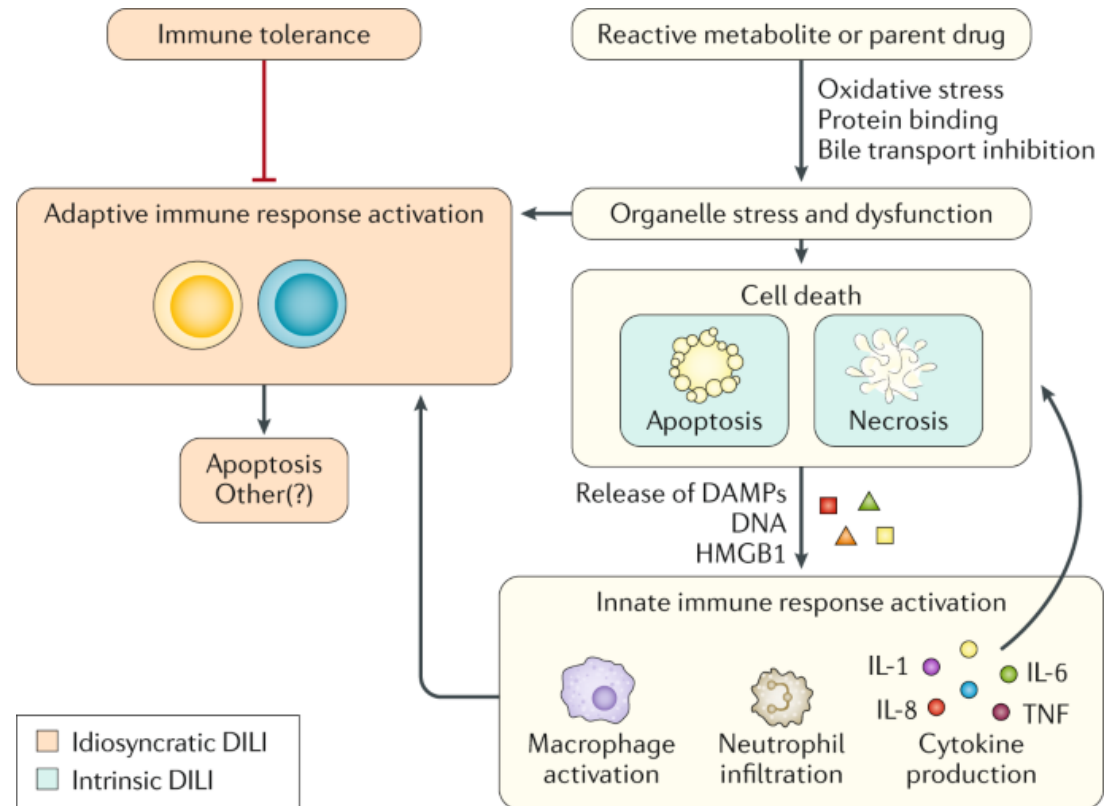


Fatty liver (microvesicular and macrovesicular steatosis)

- Alcoholic hepatitis (steatohepatitis)
- Hepatocyte swelling and necrosis
 - Mallory bodies (MB)
 - Neutrophilic reaction
 - Fibrosis (sinusoidal fibrosis pattern)

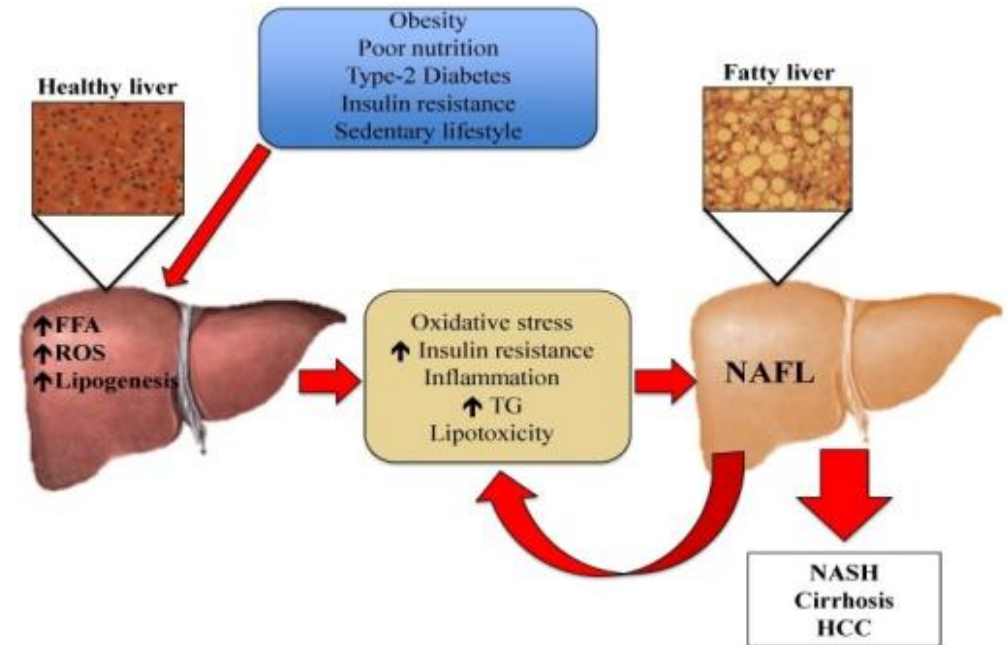
Drug induced liver injury (DILI)

- Pathogenesis:
 - dose-related intrinsic (predictable)
 - idiosyncratic (unpredictable)
- May be cholestatic or hepatocellular
- Examples of hepatic drug reactions:
 - Centrilobular necrosis: acetaminophen (paracetamol)
 - Macrovesicular steatosis: methotrexate



Metabolic dysfunction-Associated Steatotic liver Disease (MASLD)

- Very similar to liver disease caused by alcohol
- Associated with the metabolic syndrome: obesity, hyperlipidemia, hypertension and insulin resistance
- ~10-20% develop Metabolic dysfunction associated steatohepatitis (MASH) and eventually cirrhosis / hepatocarcinoma

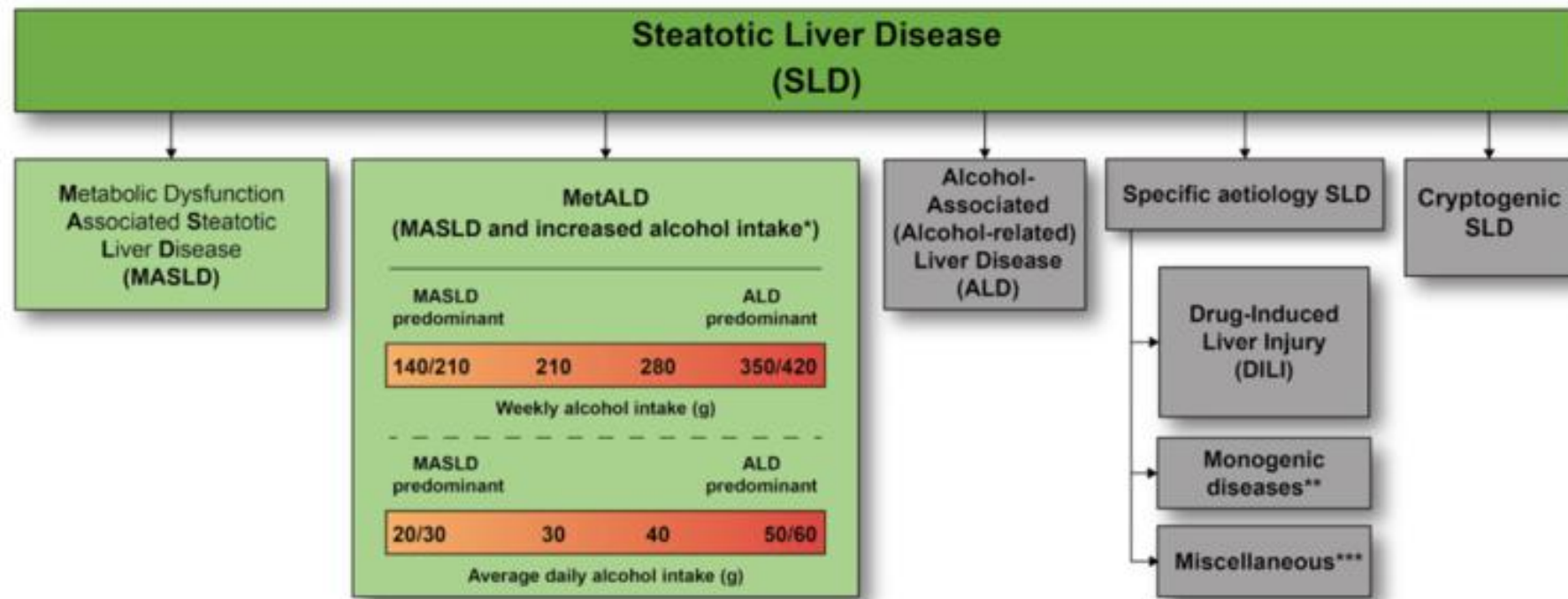


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OBS: old nomenclature!

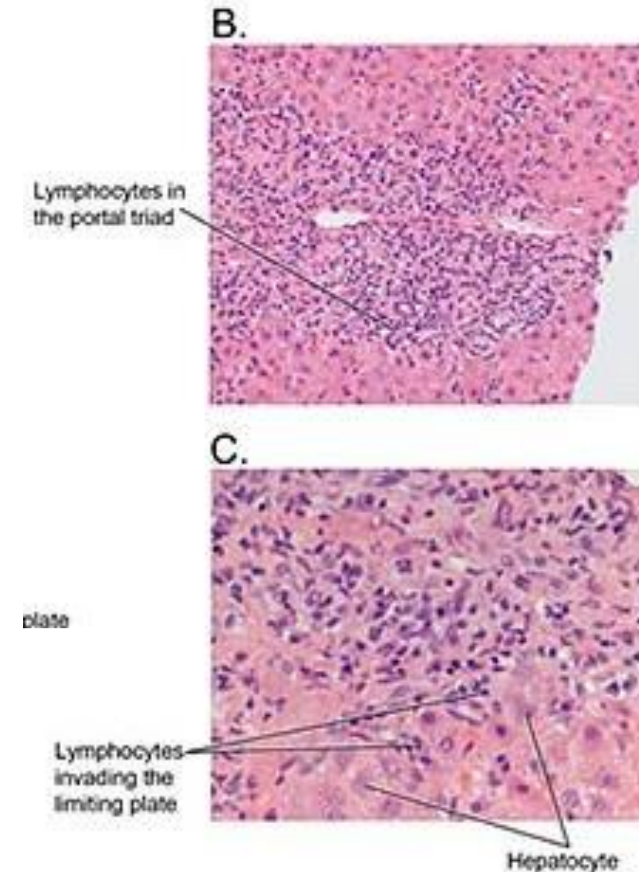
NAFLD nomenclature (2023)

Nonalcoholic fatty liver disease (NAFLD) has been changed to **metabolic dysfunction-associated steatotic liver disease-> MASLD**



Autoimmune hepatitis

- Chronic and progressive hepatitis of unknown etiology
- Autoimmune mediated injury
 - Several potential “triggers” – infection, drugs, herbal products
- Young to middle aged women primarily effected
 - M:F 1:3; ~ 40 y/o
 - Can develop a chronic hepatitis that can progress to cirrhosis
 - Acute fulminant hepatitis can occur (40%)

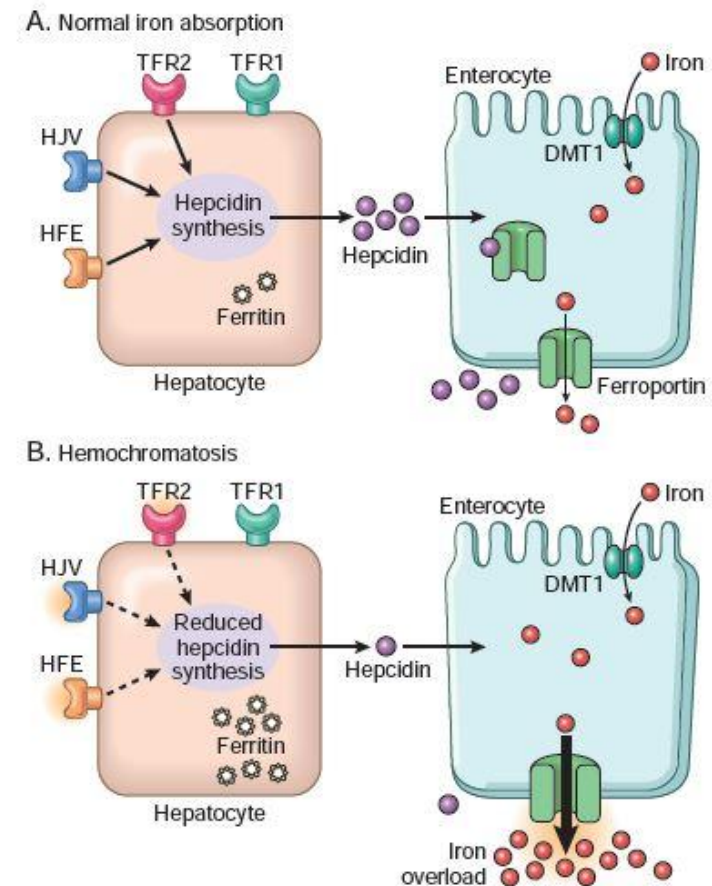


Interface hepatitis: lymphocytes in the portal triad invade the limiting plate

Hereditary metabolic liver diseases

Hemochromatosis: excessive iron accumulation in the body

- Hepcidin: master iron-regulatory hormone produced by the liver
- Hereditary hemochromatosis: mutations lower hepcidin levels or function
- Excess iron is deposited in parenchymal organs (liver, pancreas, and heart)
- ROS mediated iron toxicity causes tissue injury and promotes fibrosis



Summary: etiology of liver disease

	Acute	Chronic
Viral hepatitis	Hepatitis A, B Immune response eliminates virus	
Alcohol	Acute alcoholic hepatitis (toxic injury to hepatocytes from binge drinking)	
Metabolic dysfunction-Associated Steatotic liver Disease - MASLD		
Drugs	Toxic effect of drug on liver cells, causes necrosis (e.g. paracetamol)	
Autoimmune		
Genetic		

Summary: etiology of liver disease

	Acute	Chronic
Viral hepatitis	Hepatitis A, B Immune response eliminates virus	Hepatitis B, C Virus persists in hepatocytes, evading immune response
Alcohol	Acute alcoholic hepatitis (toxic injury to hepatocytes from binge drinking)	Steatosis alone (reversible) Steatohepatitis, cell injury and death with progressive fibrous scarring leading to cirrhosis
Metabolic dysfunction-Associated Steatotic liver Disease - MASLD		Steatosis alone (reversible) Steatohepatitis, with risk of progressive fibrosis and cirrhosis
Drugs	Toxic effect of drug on liver cells, causes necrosis (e.g. paracetamol)	Subclinical injury builds up gradually over time (e.g. methotrexate)
Autoimmune		Autoimmune hepatitis
Genetic		Haemochromatosis

Literature

- Robbins Basic Pathology, af Vinay Kumar, m.fl., 10th ed. Elsevier Saunders
- Patologi, af Niels Marcussen, Flemming B. Sørensen, Susanne Holck, Torben Steiniche og Eric Santoni-Rugiu. FADL's forlag. 2. udgave (2015).
- Or any other pathology/pathophysiology books from previous semesters

